Measures

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 66 minutes · Total marks: 66

KEY RULES

Metric	1 m = 100 cm = 1000 mm ; 1 kg = 1000 g ; 1 litre = 1000 ml
Area / volume units	$1\text{ m}^2 = 10\,000\text{ cm}^2$; $1\text{ m}^3 = 1\,000\,000\text{ cm}^3$
Speed	$\text{speed} = \dfrac{\text{distance}}{\text{time}}$
Density	$\text{density} = \dfrac{\text{mass}}{\text{volume}}$
Pressure	$P = \dfrac{F}{A}$

WORKED EXAMPLE 1

[2 marks]

Change 3.5 m^2 **into** cm^2**.**

Step 1	1 m = 100 cm, so square the factor	$1\text{ m}^2 = 100^2 = 10\,000\text{ cm}^2$
Step 2	Multiply	$3.5 \times 10\,000 = 35\,000$

Answer: $35\,000\text{ cm}^2$

WORKED EXAMPLE 2

[3 marks]

A metal block has mass 540 g and volume 200 cm^3**. Work out its density.**

Step 1	Density = mass ÷ volume	540 ÷ 200

Answer: 2.7 g/cm^3

WORKED EXAMPLE 3

[3 marks]

Change 54 **km/h into m/s.**

Step 1	Kilometres to metres	54 000 m per hour
Step 2	Hours to seconds	54 000 ÷ 3600 = 15

Answer: 15 **m/s**

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1 The exchange rate is 1 pound = 4.60 dirhams.

(a) Change 35 pounds into dirhams.

(b) Change 920 dirhams into pounds.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

2 Change 2.4 m^2 into cm^2.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 Change 3 500 000 cm^3 into m^3.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 A film starts at 21:45 and lasts 2 hours 50 minutes.

At what time does it finish?

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

5 Change 15 m/s into km/h.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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6 A metal has density 8.4 g/cm^3. Work out the mass of 250 cm^3 of the metal.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

7 A force of 600 N acts on an area of 0.5 m^2.

Work out the pressure.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

8 A plane flies from London to Accra.

The plane flies a distance of 5103 km
The flight time is 7 hours.

Work out the average speed of the plane.

..... km / h

(Total for question = 2 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2026 · Question 12

9 7 The mass of a silver coin is 48.3 g
The density of silver is 10.5 g/cm^3

Work out the volume of the silver coin.

..... cm^3

(Total for Question 7 is 2 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 7

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10 A box has a rectangular base 6 m by x m.

The force on the base is 1800 N and the pressure is 50 N/m^2.

Work out x.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

11 A box is put on a horizontal table.

The face of the box in contact with the table is a square of side 1.5 metres.
The pressure on the table due to the box is 34.8 newtons/m^2

Work out the force exerted by the box on the table.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

..... newtons

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 9

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12 A cylinder is placed on a table.

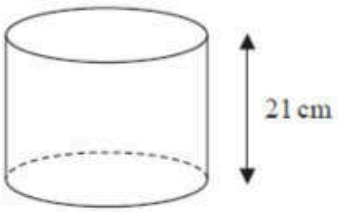


Diagram **NOT** accurately drawn

The volume of the cylinder is 1575 cm^3
The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

..... newtons/cm^2

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2024 · Question 23

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13 The diagram shows a block of iron in the shape of a cuboid.

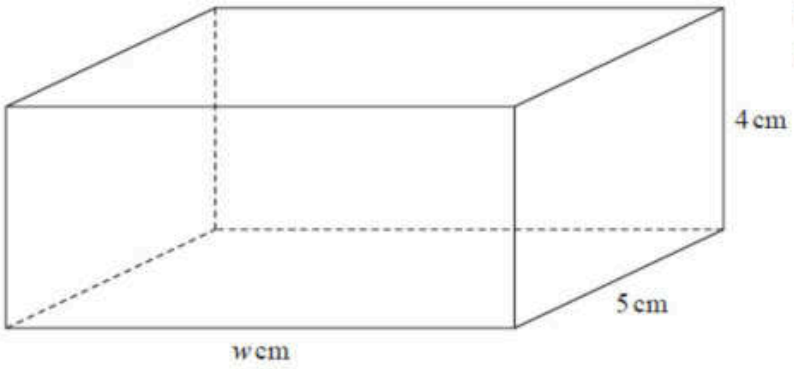


Diagram **NOT** accurately drawn

The block has length w cm, width 5 cm and height 4 cm

The density of iron is 7.8 g / cm^3
The mass of the block is 1950 g

Work out the value of w

w = ...

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2023 · Question 27

14 Henri competes in a race.

For the first part of the race Henri cycles 60 km at an average speed of 24 km/h
For the remaining part of the race Henri runs for 1 hour and 15 minutes at an average speed of 12 km/h

Work out Henri's average speed for the whole race.
Give your answer in km/h
Show your working clearly.

..... km/h

(Total for question = 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 21

15 A bus travels a distance of 1262 km from Beijing to Shanghai.
The bus takes $17\frac{1}{2}$ hours.

(a) Work out the average speed of the bus.

Give your answer, in km/h, correct to the nearest whole number.

..... km/h

(2)

(b) Change a speed of $50x$ metres per second to a speed in kilometres per hour.

..... km/h

(3)

(Total for question = 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2025 · Question 22

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16 A solid aluminium cylinder has radius 10 cm and height h cm.

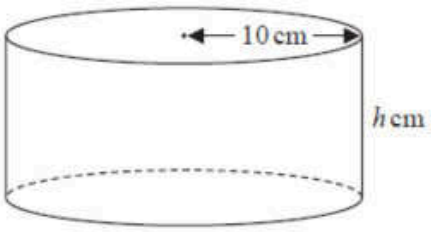


Diagram **NOT** accurately drawn

The mass of the cylinder is 5.4 kg.
The density of aluminium is 0.0027 kg/cm^3

Calculate the value of h.
Give your answer correct to one decimal place.

h = ...

(Total for question = 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2021 · Question 22

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CHALLENGE QUESTIONS

17 (a) Change 6 metres into centimetres.

..... centimetres

(1)

(b) Change 4500 grams into kilograms.

..... kilograms

(1)

Lauren has 3 litres of fruit juice.
She is going to use the fruit juice to make some drinks for a party.
Each cup of drink will contain 225 millilitres of fruit juice.

Lauren is going to make as many cups of drink as possible.

(c) Work out how much fruit juice Lauren has left when she has made as many cups of drink as possible.
Give your answer in millilitres.

..... millilitres

(4)

(Total for question = 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2022 · Question 3

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18 7 Milly went on a car journey.
She travelled from Anesey to Breigh to Clando and then to Duckbridge.

For Anesey to Breigh, Milly drove the 245 km in 2.5 hours.
For Breigh to Clando, Milly drove the 220 km at an average speed of 80 km/h
For Clando to Duckbridge, Milly drove at an average speed of 72 km/h in 50 minutes.

Work out Milly's average speed, in km/h, for the journey from Anesey to Duckbridge.
Give your answer correct to one decimal place.

..... km/h

(Total for Question 7 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2022 · Question 7

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19 7 A cylinder is placed on the ground.

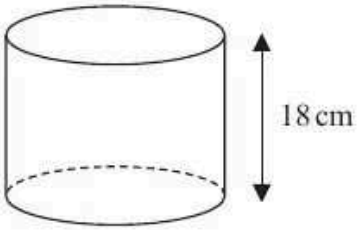


Diagram **NOT** accurately drawn

The height of the cylinder is 18 cm.

The force exerted by the cylinder on the ground is 72 newtons.
The pressure on the ground due to the cylinder is 1.4 newtons / cm^2

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cylinder.
Give your answer correct to 3 significant figures.

..... cm^3

(Total for Question 7 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 7

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20 Pedro drove from Toulouse to Montpellier in 2 hours 42 minutes.
He drove at an average speed of 90 km/hour.

Janine drove from Toulouse to Montpellier along the same route as Pedro.
The journey took her 3 hours.

Work out Janine's average speed for the journey.

..... km/hour

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 7

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21 The diagram shows a solid cube.
The cube is placed on a table so that the whole of one face of the cube is in contact with the table.

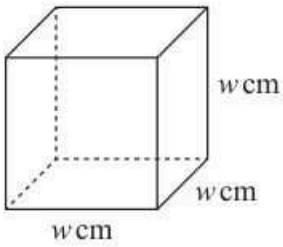


Diagram **NOT** accurately drawn

The cube exerts a force of 56 newtons on the table.
The pressure on the table due to the cube is 0.14 newtons/cm^2

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cube.

..... cm^3

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 13

TOTAL FOR PAPER IS 66 MARKS

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